

Digital Cost of Quality management in Moroccan agri-food firms: adoption insights and benefits

Ouafa BARAKAT

The National School of Business and
Management (ENCG)
Ibn Zohr University
Agadir, Morocco

<https://orcid.org/0000-0002-6134-286X>

Taib ELASSIMI

The National School of Business and
Management (ENCG)
Ibn Zohr University
Agadir, Morocco

<https://orcid.org/0009-0009-2353-8042>

Imane BOUHSAIN

The National School of Business and
Management (ENCG)
Ibn Zohr University
Agadir, Morocco

<https://orcid.org/0009-0004-7294-1202>

Abstract—This study assesses the use of digital tools in Cost of Quality (CoQ) management within Moroccan agri-food firms and the benefits they generate. A qualitative design based on semi-structured interviews with twenty quality managers and directors was analyzed thematically using MAXQDA 24. Findings reveal three adoption levels: spreadsheet-led approaches, ERP/Bi-enhanced solutions, and integrated quality suites. Reported benefits cluster into six categories: visibility and timeliness, faster CAPA cycles, improved data quality, cross-functional coordination, readiness for scaling, and a shift toward prevention. Interpreted through the Resource-Based View, these benefits emerge as digital resources—such as ERP modules, governed datasets, and integrated workflows—are transformed into organizational capabilities embedded in routines. The study provides a phased adoption pathway for managers and highlights the educational implications of integrating digital CoQ cases into curricula. To our knowledge, this is the first empirical study in the Moroccan agri-food sector linking classes of digital tools to operational benefits.

Keywords—Cost of quality; Digitalization; Quality management; Agri-food; Morocco.

I. INTRODUCTION

Today's agri-food companies face stringent quality and safety requirements while operating with perishable products and thin margins. Rapid deterioration not only threatens consumer health and satisfaction but also exposes firms to financial losses and reputational risks if not managed proactively [1], [2]. Regulatory demands and stricter private certification schemes further amplify these risks, making advanced quality management systems a business imperative even in resource-constrained contexts [3]. In this environment, quality is increasingly viewed as a strategic asset that can strengthen brand image, secure market access, and sustain competitiveness when consistently managed [4].

Within this perspective, the Cost of Quality (CoQ) provides a pragmatic framework for measuring both the costs of achieving conformance and the losses resulting from non-conformities (NC) [5]. At its core, CoQ refers to the systematic measurement of these expenditures and losses. CoQ management extends this by utilizing the data to inform resource allocation, drive continuous improvement, and support informed strategic decision-making. [6], [7]. Although several models have been proposed in the literature, ranging from activity-based costing to hidden and opportunity cost approaches, the Prevention–Appraisal–Failure (PAF) model, introduced by [8], remains the most widely applied due to its simplicity and operational relevance [5]. Despite its practical value, empirical research on CoQ management in the agri-food sector remains limited, with most studies focusing

on industrial contexts [9], [10]. In Morocco, the literature is only beginning to emerge [11], [12].

Recent advances in digital technology have reshaped quality management, moving beyond paper-based and manual approaches toward enterprise systems and analytics that enable continuous monitoring, faster detection, and more efficient corrective and preventive action (CAPA) [13]. Tools such as ERP platforms, dashboards, and integrated Quality, Health, Safety, and Environment (QHSE) traceability modules reinforce the role of quality as a foundation for broader transformation initiatives [14]. Building on this, Industry 4.0 enhances quality management with IoT sensors, SPC, machine vision, and business intelligence (BI) layers, which reduce detection-to-action time and boost traceability, especially in the agri-food sector, where perishability increases risks [15]. In Morocco, however, digital adoption in the agri-food sector is still constrained by leadership skepticism, training gaps, and weak infrastructure [16]. Beyond these barriers, adoption is also uneven across quality functions: while control and assurance benefit from sensors and automation, strategic dimensions—particularly cost analysis—remain underdeveloped [17]. This highlights a significant gap, as digital tools in CoQ management are rarely integrated into the strategic layer of food quality management.

This is likely the first empirical study on digital tools in CoQ management in the agri-food sector, focusing on Moroccan firms. The research examines digital tools and the perceived benefits associated with them. Guided by the Resource-Based View, digital tools, data, and skills are viewed as resources forming firm-specific capabilities. These routines enhance visibility, expedite corrective actions, and foster a prevention-oriented management approach. Accordingly, this study addresses two research questions:

- RQ1. Among Moroccan agri-food firms that practice CoQ management, what is the current state of digital tool adoption?
- RQ2. What benefits, both current and potential, are associated with digital CoQ management in agri-food firms?

The remainder of this paper is structured as follows: Section 2 introduces the theoretical framework, reviews prior studies, and highlights the research gap. Section 3 details the methodology. Section 4 presents the findings. Section 5 concludes with contributions, implications, limitations, and future research avenues.

II. LITERATURE REVIEW

A. Resource-Based View (RBV) perspective on digital CoQ management capabilities

The RBV [18], [19] explains sustained performance differences through resources and capabilities that are valuable, rare, inimitable, and non-substitutable (VRIN). [18] distinguishes between physical capital (including technology, equipment, location, and access to raw materials), human capital (encompassing skills, experience, and expertise), and organizational capital (comprising structures, management systems, and relational networks). Within this framework, CoQ management systems function as strategic enablers, directing investments toward technologies, skilled personnel, and integrated processes. By doing so, firms reduce poor-quality costs (such as recalls, waste, and complaints) while improving profitability, productivity, and stakeholder data sharing [18], [20]. When repeatedly enacted, these routines become embedded and harder to imitate.

In digital contexts, performance gains arise not from technology alone but from its synergy with human capital and routines. For instance, digital traceability in the agri-food sector delivers benefits when systems are integrated into daily practices and supported by competent staff [21]. Regulatory pressures may establish a baseline for data collection and disclosure; however, a strategic advantage emerges when firms tailor their digital systems to meet contextual goals and stakeholder requirements [22]. In resource-constrained environments such as Morocco's agri-food sector, advantage depends on converting fragmented tools into capabilities that integrate data pipelines, standardized PAF metrics, analytics, governance, and learning cycles—capabilities that are complex and competitor-resistant [23], [24].

RBV, in turn, clarifies what must be built for sustained impact: data and integration capabilities that stabilize pipelines from ERP/IoT; measurement capabilities that standardize PAF definitions and cost models; analytical capabilities that support dashboards, anomaly detection, and root-cause learning; and governance capabilities that assign ownership and ensure auditability [25]. This lens also explains heterogeneity in outcomes: similar tools yield different results depending on whether firms have developed VRIN-like bundles of capabilities around digital CoQ [26].

In sum, RBV clarifies how digital CoQ management can evolve into firm-specific capabilities that sustain competitiveness in agri-food contexts.

B. Prior work and research gap

Although CoQ has a solid conceptual foundation, its practical use as a strategic management tool remains limited across various sectors, with most research still at the theoretical or managerial stage [5]. This constraint is particularly evident in the agri-food industry, where empirical studies are scarce, and scholars emphasize the need for sector-specific approaches to CoQ implementation [9], [10]. In Morocco, this gap is broader: most existing studies focus mainly on non-agri-food sectors such as industry [5], [12] and automotive [20], providing little insight into how CoQ is applied within food systems, despite their vital role in exports and markets driven by certification.

In the agri-food sector, certification systems, quality management, and CoQ are interdependent. Certification schemes (e.g., BRC, IFS, GLOBALG.A.P.) provide the

external compliance frameworks that shape organizational practices [27]. Quality management systems operationalize these requirements internally, ensuring that processes and documentation meet both regulatory and buyer demands [28]. CoQ management then serves as a measurement and control mechanism within this framework, quantifying the efficiency and costs of implementing quality practices [5]. Clarifying this hierarchy situates digital CoQ as a tool that links certification pressures with strategic quality management decisions.

In parallel, a small but growing body of research explores the digitalization of CoQ management. A recent systematic review by [17] highlights that, within food quality management, the adoption of digital technologies remains uneven across the five quality functions: design, control, improvement, assurance, and strategy. While sensors and monitoring tools are widely deployed to support quality control (QC) and assurance (QA), strategic functions such as policy, planning, and cost analysis are underdeveloped, indicating that digital CoQ management at the strategic level remains largely overlooked. Complementing this perspective, recent studies outside the agri-food sector demonstrate how digitalization can enhance CoQ practices. [7], For instance, show how Industry 4.0 inspection strategies influence CoQ evaluation and failure costs using historical production data, while [29] employs simulation to compare quality-control plans in dynamic environments. Earlier work by [29] proposed a staged automation and ERP-based reporting framework, emphasizing the role of information systems in structuring CoQ data. Collectively, these studies suggest that digitalization can improve data integrity, transparency, and prevention. However, none address the specific challenges of agri-food supply chains—such as perishability, dispersed production locations [1], certification requirements, and varying levels of digital literacy—nor the unique context of Morocco, underscoring the need for empirical research.

III. METHODOLOGY

This study employs a qualitative research design to explore the use of digital tools in CoQ management within Moroccan agri-food firms and the benefits they yield, aligning with an interpretive paradigm that emphasizes understanding stakeholder perceptions and contextual nuances [30]. Qualitative methods are particularly well-suited for examining digital integration in developing economies, providing in-depth insights into this underexplored area [31].

Participants were selected using a snowball sampling strategy, suitable for reaching specialized populations in quality management [32]. Initial contacts were established through professional networks, and subsequent referrals expanded the sample until theoretical saturation was achieved, defined as the point at which no new themes emerged [33].

Indeed, qualitative studies frequently employ smaller, context-specific samples to generate rich insights rather than aiming for broader generalizability [34], [35]. The homogeneity of this sample justified achieving data saturation, aligning with methodological recommendations that suggest smaller qualitative samples can provide sufficient depth and reliability when the participant group is homogeneous and consists of information-rich individuals [36], [37].

The final sample consisted of 20 quality managers and directors from diverse agri-food companies, representing both SMEs and larger firms in key regions.

Data were collected through semi-structured interviews, providing flexibility to explore emerging insights while focusing on the research questions [38]. Each interview lasted 45–60 minutes and was conducted either virtually or in person between December 2024 and May 2025 in French. The interviews were audio-recorded with consent. It covered topics on RQ1 and RQ2, including digital tools and perceived benefits in CoQ management.

All interviews were recorded, transcribed, and analyzed in MAXQDA 24, which supports systematic coding and thematic analysis [39]. We adopted a grounded-theory–informed approach. During open coding, we read transcripts line-by-line and assigned short labels close to participants’ words (e.g., adoption of CoQ or not, tools used, coordination routines, perceived improvements) [40]. In axial coding, we grouped similar codes into a small set of categories related to Digital CoQ—specifically, clusters for tool/level of adoption and benefits for the adopters. [33]. In selective coding, we integrated the categories and codes developed during open and axial coding [41]. Through selective coding, we identified links among categories and constructed an overarching narrative of the dataset, enabling coherent insights and a comprehensive understanding of underlying meanings [42]. Following [43], data from several examples were organized into tables (MAXQDA’s Segment Matrix) to facilitate comparisons and synthesis. The code relations browser was applied to check co-occurrence patterns (e.g., between digital tools and perceived benefits), while summary grids were generated to consolidate interpretations per case. Intra- and inter-case analyses followed [44] guidelines, ensuring the transparency and reliability of the findings. To enhance trustworthiness, two researchers double-coded an initial subset, reconciled discrepancies through discussion, and refined the codebook before coding the remaining transcripts. Member checking with selected informants was used to validate thematic interpretations.

To strengthen rigor, member checking was conducted by sharing transcript summaries with participants, and peer debriefing was carried out among the research team [45]. Ethical protocols included obtaining informed consent, maintaining confidentiality, and anonymizing participants, who were identified by labels R1 through R20. Table I provides descriptive information on participant codes, roles, years of experience, and CoQ measurement.

TABLE I. PARTICIPANTS’ PROFILE

Case Code	Position	Experience (in years)	CoQ practice
R1	Quality Director	12	Yes
R2	Quality Manager	5	No
R3	QHSE Manager	5	Yes
R4	Quality Manager	5	Yes
R5	Quality Manager	5	No
R6	QHSE Manager	18	Yes
R7	QHSE Manager	9	Yes
R8	QHSE Manager	8	Yes
R9	Quality Director	10	No
R10	QHSE Manager	25	Yes
R11	Quality Manager	9	No
R12	QHSE Manager	9	Yes
R13	Quality Manager	5	No
R14	Quality Manager	6	No
R15	QHSE Manager	10	Yes
R16	Quality Director	12	Yes

R17	Quality Director	9	No
R18	Quality Director	10	Yes
R19	Quality Manager	12	No
R20	Quality Director	12	No

IV. FINDINGS

Across the sample, quality management systems are driven mainly by certification and involve multiple schemes. The most common standards are BRC (British Retail Consortium) (70%), GLOBAL G.A.P. (Good Agricultural Practices) (50%), SMETA (SEDEX Members Ethical Trade Audit) (45%), GRASP (Risk Assessment on Social Practice) (45%), IFS (International Featured Standards) (30%), and ISO 22000 (25%). Other schemes appear less frequently, such as SPRING (Sustainable Program for Irrigation and Groundwater Use) (25%), HALAL (20%), LEAF (Linking Environment and Farming) (20%), ISO 9001 (15%), KOSHER (15%), Chain of Custody (15%), and various niche or buyer-specific programs at 5% each. Multi-certification is a standard practice: 90% of firms hold more than one certificate [27].

Certification adoption is predominantly buyer-led: 85% report implementing schemes due to customer requirements, while 15% adopted them voluntarily. Reported advantages converge on seven consistent themes: (i) access to international/premium markets and smoother acceptance by EU/UK/US retailers; (ii) structuring and standardization of processes and documentation; (iii) stronger food-safety and traceability across the chain; (iv) risk reduction with fewer non-conformities and complaints; (v) enhanced credibility and brand image with customers, partners, and authorities; (vi) deeper employee engagement and quality culture through audits and training; and (vii) improved readiness for audits, tenders, and compliance—consistent with the view that certification acts both as a market license and an internal management scaffold in export-oriented agri-food chains noted in recent Quality management discussions.

Formal CoQ practice was observed in 55% of firms, while 45% reported no CoQ measurement. This proportion is higher than the national average of 35.5% reported by [5], suggesting an increasing awareness of the benefits of systematic oversight and management of quality-related costs.

A. Current state of digital tool adoption among CoQ-practicing firms

Across the 11 CoQ-practicing firms, digital adoption clusters into three category levels (Table II):

TABLE II. LEVELS OF DIGITAL TOOL ADOPTION IN CoQ MANAGEMENT AMONG MOROCCAN AGRI-FOOD FIRMS (CoQ-PRACTICING SAMPLE ONLY)

Category /Share	Typical tools & data flow	Participant’s quotes
Level 1 Spreadsheet-led CoQ (≈64%)	Excel (incl. shared drives), Word/PDF forms; paper NC logs; manual roll-ups to PAF; CAPA follow-up tracked in shared files/email; limited audit trail; weak	“Our CoQ relies mainly on Excel and a documentary QMS. New technologies are not yet fully integrated, but we see the potential to automate tracking, analyze faster, and reduce input errors.” (R1) “Like many in Morocco’s agri-food sector, we still work with simple tools—Excel and paper—which limits tracking precision; cost, lack of training, and operational urgencies slow the transition.” (R3) “Digital tools help us trace non-conformities, centralize quality data,

	master-data governance.	and react faster... we use shared Excel files, which eases information flow between farm, packhouse, and QA." (R4) "We use Excel, Drive, Word. The technological level remains basic, which limits automation of cost tracking. Digitalizing quality/CoQ PAF indicators is necessary to go further." (R6) "Paper QMS; only Excel is used for CoQ." (R7) "We log NCs and CAPA in shared Excel; follow-up is manual and depends on the supervisor." (R7) "We use structured Excel to track cost indicators (scrap, losses, complaints) ... imperfect, but it already lets us trigger corrective decisions." (R15) "We are still semi-manual; planned steps include digitalizing quality records (NCs, scrap, indicators), real-time anomaly tracking via connected tools, and automated Excel dashboards to link CoQ with operations." (R16)
Level 2 ERP/BI enhanced CoQ (≈27%)	ERP with quality/cost KPIs; Power BI/VBA dashboards; some in-line QC/traceability automation; periodic data refresh. QMS/ERP for NC/CAPA workflow;	"ERP deployment is underway to track costs by work center; the objective is to digitalize all quality records and automate CoQ extractions." (R8) "NCs are opened in the QMS, and we track CAPA to closure" (R8) "Technologies are starting to matter—automation of traceability and in-line QC, analysis via ERP or specialized industry tools with CoQ KPIs, and visual dashboards." (R10) "Automated dashboards with alerts when thresholds are crossed; advanced analytics (VBA/Power BI)" (R18) "a project to centralize quality-cost data in a single decision tool." (R18)
Level 3 Integrated QHSE (≈9%)	QHSE digital software with work flowed NC/CAPA; cost capture at source (NC ↔ cost center/work order); governed master data;	"We monitor non-conformities in real time, automate quality data collection, and run statistical root-cause analysis via QHSE software; digital traceability and integrated document management generate alerts when drifts occur." (R12) "Once we linked NC categories to cost centers, PAF trends are generated automatically for monthly reviews." (R12) "CAPA deadlines and ownership are enforced in the system." (R12)

A progressively structured workflow underpins CoQ data management across levels. At the basic level, firms capture non-conformities in spreadsheets or paper forms and later consolidate them into PAF reports. Intermediate adopters add ERP modules and BI dashboards, enabling semi-automated validation, classification, and aggregation of quality-cost data. Advanced adopters institutionalize these practices via embedded NC→CAPA workflows, governed master data, scheduled dashboards, and centralized decision support. Despite differences in tool sophistication, a typical sequence emerges—NC capture → validation → aggregation into PAF → management review—with increasing degrees of automation, integration, and decision support.

Answer to RQ1. Among Moroccan agri-food firms that practice Cost-of-Quality (CoQ) management, what is the current state of digital tool adoption?

Based on Table 2 (CoQ-practicing subsample, $n = 11$), digital maturity clusters into three tiers: Spreadsheet-led (≈64%), ERP/BI-enhanced (≈27%), and Integrated QHSE (≈9%). In most firms, digital tools still function mainly as repositories: PAF is manually compiled and CAPA follow-up occurs outside core systems; even hybrid sites that add QMS/ERP modules and dashboards frequently export to Excel for cost assignment and monthly aggregation—reflecting heterogeneous costing systems and inconsistent CoQ structures that limit cross-firm comparability[46]. Viewed through RBV, only the integrated tier (Level 3) exhibits VRIN-like capability bundles—governed data pipelines, standardized CoQ/PAF metrics, embedded analytical routines—capable of sustaining advantages under volatility. The predominance of Level-1 adoption in a resource-constrained, multi-site agri-food context indicates headroom for building these bundles; targeted investment in interoperability and governance should therefore accelerate progress toward integration. In practical terms, automation and digitalization of CoQ are more effective when implemented together with ERP in a shared information environment, rather than as stand-alone tools [29]. By contrast, studies in industrialized contexts (e.g., automotive and electronics) reveal higher penetration of ERP-embedded CoQ systems and an early shift toward predictive analytics, confirming that sectoral and regional constraints influence digital maturity paths [47]. This supports the early Quality 4.0 adoption sequence—digitalize, integrate (with ERP/QMS), then predict/optimize—and explains why our single integrated case (NC ↔ cost center linkages, governed master data, scheduled dashboards) yields the most consistent management review results[48].

B. Benefits and opportunities that firms perceive from digital CoQ management

To synthesize both realized gains and anticipated advantages across digital maturity levels, Table III groups the benefits and opportunities of digital CoQ management by experienced (Levels 2–3) versus perceived (Level 1) outcomes, with representative participant quotes.

TABLE III. BENEFITS AND OPPORTUNITIES THAT FIRMS PERCEIVE FROM DIGITAL CoQ MANAGEMENT

Benefit category	Adopter level	Participant's quotes
Visibility & timeliness	Experienced (Level 2–3)	"Automated dashboards send alerts when thresholds are exceeded, which improves responsiveness." (R18) "New technologies provide visual dashboards and analytical views that enhance performance monitoring." (R10) "Real-time tracking of non-conformities helps us detect issues sooner and coordinate a response." (R12)
	Perceived (Level 1)	"If we had digital dashboards, we could improve the visibility of non-conformities and respond faster." (R6) "If we moved beyond shared Excel files, centralized digital tools would speed up escalation." (R4) "If new technologies were integrated, tracking could be automated and analysis faster." (R1)

Faster CAPA cycles	Experienced (Level 2–3)	“Real-time alerts on non-conformities let us trigger corrective actions immediately when deviations occur.” (R12) “Dashboards shorten the loop between detection and action in our CAPA process.” (R18)
	Perceived (Level 1)	“If we had digital monitoring of cost items, we could take corrective decisions more quickly.” (R15) “With digital tools, CAPA lead times would be reduced compared to our current manual process.” (R3)
Data quality & error reduction	Experienced (Level 2–3)	“Automating traceability and in-line QC reduces manual recording and improves data reliability.” (R10) “Integrated document management limits duplication and cuts reporting errors.” (R12)
	Perceived (Level 1)	“If we automated data entry, manual errors would drop and analysis would be faster.” (R1) “Digitizing CoQ indicators would make reporting more reliable and reduce mistakes.” (R6)
Cross-functional coordination	Experienced (Level 2–3)	“ERP deployment gives us a common view of costs by workstation, which improves coordination between production and quality.” (R8) “Shared dashboards align teams and make follow-up easier across departments.” (R18)
	Perceived (Level 1)	“If we had an integrated platform, information flow between the farm, the packhouse, and QA would be much smoother.” (R4) “A digital space to centralize quality data would improve collaboration across teams.” (R15)
Readiness for integration & scaling	Experienced (Level 2–3)	“With ERP deployment, we can now track CoQ by workstation, to digitize all quality records.” (R8) “Power BI or ERP modules could standardize and secure CoQ calculations across sites.” (R16)
	Perceived (Level 1)	“If we had interoperable tools, we could scale the same CoQ indicators across units instead of rebuilding spreadsheets.” (R6) “If integration were in place, consolidation across sites would be straightforward.” (R1)
Shift toward prevention-oriented management.	Experienced (Level 2–3)	“Automating traceability and in-line quality control enables earlier detection and supports preventive action.” (R10) “Real-time monitoring and integrated documentation help us move from corrective to preventive approaches.” (R12)
	Perceived (Level 1)	“If early-warning indicators were digital, we could act before failures instead of reacting after the fact.” (R3) “If we digitized key indicators, prevention would become part of the routine, not an exception.” (R6)

Across cases, Level 2–3 firms report tangible improvements in visibility, faster CAPA cycles, data reliability, and cross-functional coordination, reflecting the effect of workflow-based NC→CAPA and dashboard-driven reviews. By contrast, Level 1 firms consistently identify the same gains as near-term opportunities, contingent on moving beyond shared spreadsheets toward interoperable tools and governed data. Readiness for scaling—via standardized PAF logic and multi-site roll-ups—and a stronger prevention orientation, including earlier detection and targeted actions, both increase with digital maturity, indicating that the value of digitization intensifies as integration deepens.

Answer to RQ2. What benefits, both current and potential, are associated with digital CoQ management in agri-food firms?

Drawing on Table 3, CoQ-practicing firms converge on the same set of benefits—realized at higher maturity (Levels 2–3) and anticipated at lower maturity (Level 1): clearer and timelier visibility, shorter CAPA loops, fewer reporting errors, better cross-functional alignment, easier consolidation across sites, and a shift toward prevention. The emphasis on prevention is consistent with the economic intuition that upstream investment delivers the most significant leverage (the 1–10–100 rule) [46]. Reported effects also align with evidence that digital tools reduce waste, lower non-quality costs, and support sustainability outcomes in agri-food operations [49], [50].

Through the RBV lens, these benefits arise as firms orchestrate ERP/BI, integration platforms, and governed data into firm-specific capability bundles that enhance visibility and resilience under volatile and uncertain business conditions [21], [51]. In certification-intensive Moroccan agri-food chains, the same capability building can extend to green innovation and footprint reduction when digital resources are bundled with routines and KPIs [52]. In summary, the data suggest that even incremental adoption yields meaningful operational gains [49], [50]. Scaling these gains—standardized PAF, predictive prevention, and multi-site coordination—depends on deeper integration and governance consistent with the Level-2→Level-3 transition.

These outcomes depend on digital tools rooted in CoQ workflows (NC capture → analysis → CAPA) and trained staff: “Our dashboards only prompt action because each alert opens a CAPA with an owner; training line leaders made the loop stick” (R12). Sector specifics—spread-out sites, varied digital skills, and budget limits—mean scaling needs technical integration and organizational change: “With farms dispersed, uneven digital skills, and tight SME budgets, we phased integration and adapted routines to keep pace” (R18). Several teams also describe a strategic shift even with partial digitalization: “Using shared spreadsheets and simple alerts, seeing costs at source daily shifted us from firefighting to preventive fixes” (R3).

Positioning within digital-maturity frameworks. Our three adoption tiers align with peer-reviewed maturity logics. In smart-manufacturing maturity models, firms evolve from basic computerization and isolated data toward connected, standardized flows and, ultimately, predictive/adaptive capabilities [53], [54]. Quality 4.0 scholarship likewise argues that performance gains arise when data, connectivity, and analytics are embedded in quality governance and skills—an

especially salient lens for food chains [55], [56]. Consistent with these frameworks, L1 (spreadsheet-led) reflects low integration and manual handoffs yielding primarily descriptive visibility [53]; L2 (ERP/B1-enhanced) corresponds to mid-maturity states emphasizing integrated data models, cross-functional KPIs, and diagnostic/near-real-time visibility [54]; and L3 (integrated QHSE suites) aligns with higher maturity featuring workflow automation, closed-loop CAPA, audit-ready traceability, and forward-looking analytics [55], [56]. This mapping explains the step-changes we observe from L1 → L2 → L3: benefits shift from error reduction and basic visibility toward coordinated, timely action and preventive, system-level improvement [54], [55].

V. CONCLUSION

This paper documents the current state of digital tool adoption in CoQ management within Moroccan agri-food firms and the outcomes reported by adopters. Evidence indicates three maturity tiers—spreadsheet-led routines, ERP/B1-enhanced practices, and integrated QHSE systems—with the majority of firms still relying on spreadsheets. Where integration deepens, managers describe clearer and timelier visibility of PAF indicators, shorter CAPA cycles, fewer reporting errors, stronger coordination across departments, better readiness for multi-site consolidation, and a gradual reorientation toward prevention. The findings, therefore, position digital tools not as repositories but as enablers of actionable oversight when they are connected to PAF measurement and everyday decision routines.

The results suggest that digital adoption of CoQ management should be pursued as a progressive capability-building process rather than a single technological investment. Managers can start by stabilizing master data and introducing simple dashboards to track PAF indicators, then move toward ERP or integrated QHSE systems that automate non-conformity capture, cost assignment, and CAPA workflows. Even modest tools can enhance visibility, minimize reporting errors, and expedite CAPA cycles, while advanced integration facilitates cross-functional coordination and multi-site consolidation. For Moroccan agri-food firms, where certification is both a market entry condition and a reputational safeguard, positioning digital CoQ as a strategic enabler helps reduce waste, strengthen buyer confidence, and build resilience under resource constraints.

For education, the findings emphasize the importance of integrating digital CoQ management into curricula in quality management, operations, and agribusiness programs. Traditional teaching on the PAF model and quality costing can be enriched with modules on ERP/B1 dashboards, QHSE integration, and case studies from export-oriented food chains. Exposing students to simulation exercises, digital tool demonstrations, and cross-functional cost analysis prepares them to treat CoQ not only as a cost-tracking mechanism but also as a strategic capability. Embedding these elements into MBA, engineering, and agri-food programs equips future managers with the skills to navigate certification pressures, sustainability requirements, and digital transformation in practice.

This qualitative study has some limitations. It examined 20 Moroccan agri-food companies, with only 11 practicing CoQ, providing a detailed context but limiting broader applicability. The evidence is based on self-reports, which may be biased; however, measures such as member checking

and reflexive memos were employed to increase trust. Its cross-sectional approach captures practices at a single point in time, not over periods. Moreover, the benefits identified are presented in declarative terms without quantitative measurement of cost savings, productivity gains, or defect reduction. Future studies should supplement these perceptions with audited financial and operational data. Without a validated digital-CoQ maturity scale and audited cost data, maturity levels are only observed patterns rather than standardized measures. Therefore, future research should develop a validated maturity assessment tool, combine qualitative insights with quantitative indicators, and use longitudinal or mixed-methods approaches to deepen understanding. Comparing different sectors, regions, and regulations can enhance external validity. It is also important to identify obstacles and enablers related to adoption and to create phased digitalization roadmaps tailored to resource-constrained firms.

ACKNOWLEDGMENT (Heading 5)

The authors thank the participating agri-food firms and quality managers for their valuable contributions.

REFERENCES

- [1] S. Khazaeli, R. Kalvandi, et H. Sahebi, « A multi-level multi-product supply chain network design of vegetables products considering costs of quality: A case study », *PLOS ONE*, vol. 19, n° 9, Art. n° 9, sept. 2024, doi: 10.1371/journal.pone.0303054.
- [2] T. Tambunan, « A conceptual framework of quality cost chain in strategic cost management », *The TQM Journal*, vol. 36, n° 3, Art. n° 3, 2024, doi: 10.1108/TQM-09-2021-0281.
- [3] P. Spagnoli, L. Defalchidu, P. Vlerick, et L. Jacxsens, « The Relationship between Food Safety Culture Maturity and Cost of Quality: An Empirical Pilot Study in the Food Industry », *Foods*, vol. 13, n° 4, Art. n° 4, 2024, doi: 10.3390/foods13040571.
- [4] A. Sailaja, P. C. Basak, et K. G. Viswanadhan, « Hybrid fuzzy MCDM model for effective utilisation of quality cost analysis in manufacturing firms », *Int. J. Prod. Qual. Manage.*, vol. 24, n° 2, p. 219-241, 2018, doi: 10.1504/IJPM.2018.091793.
- [5] L. Ayach, A. Anouar, et M. Bouzziri, « Quality cost management in Moroccan industrial companies: Empirical study », *Journal of Industrial Engineering and Management*, vol. 12, p. 97, févr. 2019, doi: 10.3926/jiem.2749.
- [6] K. K. Castillo-Villar, N. R. Smith, et J. L. Simontoney, « The impact of the cost of quality on serial supply-chain network design », *Int J Prod Res*, vol. 50, n° 19, p. 5544-5566, 2012, doi: 10.1080/00207543.2011.649802.
- [7] A. M. Reis, A. Dall-Orsoletta, E. Nunes, L. Costa, et S. Sousa, « Quality costs and Industry 4.0: inspection strategy modelling and reviewing », *Int J Adv Manuf Technol*, p. 1-15, févr. 2024, doi: 10.1007/s00170-024-13184-9.
- [8] A. V. Feigenbaum, *TQC—Total Quality Control*. Harvard Business Review, 1956.
- [9] C. Dimitrantzou, E. Psomas, et F. Vouzas, « Future research avenues of cost of quality: a systematic literature review », *The TQM Journal*, vol. 32, n° 6,

- Art. n° 6, janv. 2020, doi: 10.1108/TQM-09-2019-0224.
- [10] E. Psomas, C. Dimitrantzou, et F. Vouzas, « Practical implications of cost of quality: a systematic literature review », *Int. J. Product. Perform. Manage.*, vol. 71, n° 8, Art. n° 8, 2022, doi: 10.1108/IJPPM-10-2020-0524.
- [11] W. Jabri, L. Ayach, et K. Moustaid, « Bibliographical Review About Quality Cost Management Systems in the Moroccan Market », *International Journal of Supply Chain Management*, vol. 11, n° 4, Art. n° 4, août 2022, doi: 10.59160/ijscm.v11i4.6084.
- [12] W. Jabri, K. Azoulay, et K. Moustaid, « Quality Cost Management in Moroccan Industries: Challenges and Sustainable Solutions », in *Technical Innovation and Modeling in the Biological Sciences*, IGI Global Scientific Publishing, 2025, p. 397-406. doi: 10.4018/979-8-3693-9450-2.ch021.
- [13] B. Lei, « Research on the Impact of Digital Transformation on Enterprise Operational Efficiency », *Frontiers in Business, Economics and Management*, vol. 18, n° 1, Art. n° 1, janv. 2025, doi: 10.54097/avegqh39.
- [14] A. M. Dias, A. M. Carvalho, et P. Sampaio, « Quality 4.0: literature review analysis, definition and impacts of the digital transformation process on quality », *International Journal of Quality & Reliability Management*, vol. 39, n° 6, p. 1312-1335, 2022, doi: 10.1108/IJQRM-07-2021-0247.
- [15] E. E. Broday, « The evolution of quality: from inspection to quality 4.0 », *International Journal of Quality and Service Sciences*, vol. 14, n° 3, p. 368-382, janv. 2022, doi: 10.1108/IJQSS-09-2021-0121.
- [16] M. El Feniche *et al.*, « Perceptions Regarding Quality Management Systems of Moroccan Medical Laboratory Practitioners: A Cross-sectional Study », 2024, doi: 10.2174/0118749445363860241205063519.
- [17] F. A. P. Peres, B. A. Bondarczuk, L. de C. Gomes, L. de C. Jardim, R. G. de F. Corrêa, et I. C. Baierle, « Advances in Food Quality Management Driven by Industry 4.0: A Systematic Review-Based Framework », *Foods*, vol. 14, n° 14, p. 2429, janv. 2025, doi: 10.3390/foods14142429.
- [18] J. Barney, « Firm Resources and Sustained Competitive Advantage », *Journal of Management*, vol. 17, n° 1, p. 99-120, mars 1991, doi: 10.1177/014920639101700108.
- [19] E. T. Penrose, *The Theory of the Growth of the Firm*, 2nd ed. Basil Blackwell, 1959.
- [20] N. Sraidi et C. Bakour, « Étude qualitative exploratoire sur l'impact de la gestion coûts d'obtention de la qualité sur la performance financière des entreprises de l'industrie automobile au Maroc », vol. 4, n° 6, 2023.
- [21] G. M. Razak, L. C. Hendry, et M. Stevenson, « Supply chain traceability: a review of the benefits and its relationship with supply chain resilience », *Production Planning & Control*, vol. 34, n° 11, p. 1114-1134, août 2023, doi: 10.1080/09537287.2021.1983661.
- [22] M. Montecchi, K. Plangger, et D. C. West, « Supply chain transparency: A bibliometric review and research agenda », *International Journal of Production Economics*, vol. 238, p. 108152, août 2021, doi: 10.1016/j.ijpe.2021.108152.
- [23] A. Benayad *et al.*, « Transition toward Sustainability in the Moroccan Food System: Drivers, Outcomes, and Challenges », *World*, vol. 5, n° 3, Art. n° 3, sept. 2024, doi: 10.3390/world5030032.
- [24] F. M. B. Epelbaum et M. G. Martinez, « The technological evolution of food traceability systems and their impact on firm sustainable performance: A RBV approach », *Int. J. Prod. Econ.*, vol. 150, p. 215-224, avr. 2014, doi: 10.1016/j.ijpe.2014.01.007.
- [25] C. Ni, C. Zhang, W. Lu, H. Wang, et J. Wu, « Enabling Intelligent Decision Making and Optimization in Enterprises through Data Pipelines », *World Journal of Innovation and Modern Technology*, vol. 7, n° 2, p. 99-107, mars 2024, doi: 10.53469/wjimt.2024.07(02).13.
- [26] T. Atobishi, S. Moh'd Abu Bakir, et S. Nosratabadi, « How Do Digital Capabilities Affect Organizational Performance in the Public Sector? The Mediating Role of the Organizational Agility », *Administrative Sciences*, vol. 14, n° 2, p. 37, févr. 2024, doi: 10.3390/admsci14020037.
- [27] K. Latouche et E. Chevassus-Lozza, « Retailer Supply Chain and Market Access: Evidence From French Agri-food Firms Certified with Private Standards », *The World Economy*, vol. 38, n° 8, p. 1312-1334, 2015, doi: 10.1111/twec.12191.
- [28] T. Bar et Y. Zheng, « Choosing Certifiers: Evidence from the British Retail Consortium Food Safety Standard », *American Journal of Agricultural Economics*, vol. 101, n° 1, p. 74-88, 2019, doi: 10.1093/ajae/aay024.
- [29] O. Maslak, V. Danylko, et M. Skliar, « Automation and Digitalization of Quality Cost Management of Power Engineering Enterprises », in *2020 IEEE Problems of Automated Electrodrive. Theory and Practice (PAEP)*, sept. 2020, p. 1-4. doi: 10.1109/PAEP49887.2020.9240876.
- [30] J. W. Creswell et C. N. Poth, *Qualitative inquiry and research design: Choosing among five approaches*. Sage publications, 2016.
- [31] R. K. Yin, *Case Study Research and Applications: Design and Methods*. Los Angeles: SAGE Publications, Inc, 2018.
- [32] M. Q. Patton, *Qualitative Research & Evaluation Methods: Integrating Theory and Practice*. Thousand Oaks, California: SAGE Publications, Inc, 2015.
- [33] A. Strauss et J. Corbin, *Basics of qualitative research*. Sage publications, 1990. Consulté le: 8 janvier 2024. [En ligne]. Disponible sur: <https://genderopen-develop.ub.hu-berlin.de/bitstream/handle/25595/12/whatsnew7.pdf?sequence=1>
- [34] V. Braun et V. and Clarke, « To saturate or not to saturate? Questioning data saturation as a useful concept for thematic analysis and sample-size rationales », *Qualitative Research in Sport, Exercise and Health*, vol. 13, n° 2, p. 201-216, mars 2021, doi: 10.1080/2159676X.2019.1704846.
- [35] K. Malterud, V. D. Siersma, et A. D. Guassora, « Sample Size in Qualitative Interview Studies:

- Guided by Information Power », *Qual Health Res*, vol. 26, n° 13, p. 1753-1760, nov. 2016, doi: 10.1177/1049732315617444.
- [36] C. R. Boddy, « Sample size for qualitative research », *Qualitative Market Research: An International Journal*, vol. 19, n° 4, p. 426-432, sept. 2016, doi: 10.1108/QMR-06-2016-0053.
- [37] C. S. Constantinou, M. Georgiou, et M. Perdikiogianni, « A comparative method for themes saturation (CoMeTS) in qualitative interviews », *Qualitative Research*, vol. 17, n° 5, p. 571-588, oct. 2017, doi: 10.1177/1468794116686650.
- [38] S. Brinkmann et S. Kvale, *InterViews: Learning the Craft of Qualitative Research Interviewing*. Los Angeles London New Delhi Singapore Washington DC: SAGE Publications, Inc, 2015.
- [39] U. Kuckartz et S. Rädiker, *Analyzing Qualitative Data with MAXQDA: Text, Audio, and Video*. Cham: Springer International Publishing, 2019. doi: 10.1007/978-3-030-15671-8.
- [40] B. Glaser et A. Strauss, *Discovery of Grounded Theory: Strategies for Qualitative Research*. New York: Routledge, 2017. doi: 10.4324/9780203793206.
- [41] M. A. Cascio, E. Lee, N. Vaudrin, et D. A. Freedman, « A Team-based Approach to Open Coding: Considerations for Creating Inter-coder Consensus », *Field Methods*, vol. 31, n° 2, p. 116-130, mai 2019, doi: 10.1177/1525822X19838237.
- [42] L. D. Goodwin, « Interrater Agreement and Reliability », *Measurement in Physical Education and Exercise Science*, vol. 5, n° 1, p. 13-34, mars 2001, doi: 10.1207/S15327841MPEE0501_2.
- [43] M. B. Miles et A. M. Huberman, *Qualitative data analysis: An expanded sourcebook*. sage, 1994.
- [44] R. K. Yin, *Case study research: Design and methods*, vol. 5. sage, 2009.
- [45] N. K. Denzin et Y. S. Lincoln, *The SAGE Handbook of Qualitative Research*. SAGE, 2011.
- [46] N. Sharma et B. Laishram, « Decoding the Emergent Patterns of Cost of Quality through the Lens of Sociotechnical Systems: A Bibliometric Analysis », *Journal of Construction Engineering and Management*, vol. 151, n° 4, p. 03125002, avr. 2025, doi: 10.1061/JCEMD4.COENG-15023.
- [47] S. D. Sousa, L. S. Dias, et E. P. Nunes, « Assessment of Alternative Quality Control Plans in Dynamic Contexts: A Simulation Approach », in *Lect. Notes Mech. Eng.*, Springer Science and Business Media Deutschland GmbH, 2024, p. 443-450. doi: 10.1007/978-3-031-38165-2_52.
- [48] J. Antony, V. Swarnakar, M. Sony, O. McDermott, et R. Jayaraman, « How do organizational performances vary between early adopters and late adopters of Quality 4.0? An exploratory qualitative study », *The TQM Journal*, vol. ahead-of-print, n° ahead-of-print, janv. 2023, doi: 10.1108/TQM-03-2023-0083.
- [49] K. Azzaoui, S. Yousfi, et M. L. Bouamrani, « The Benefits of Combining Digitalization with Quality Tools: Application in the Field of Wiring Systems Manufacturing for the Automotive Industry », in *Smart Applications and Data Analysis*, M. Hamlich, L. Bellatreche, A. Siadat, et S. Ventura, Éd., Cham: Springer International Publishing, 2022, p. 308-322. doi: 10.1007/978-3-031-20490-6_25.
- [50] L. M. Idigova, S. A. Plis, et H. G. Chaplaev, « Quality Management Digitalization », *European Proceedings of Social and Behavioural Sciences*, vol. Social and Cultural Transformations in the Context of Modern Globalism (SCTCMG 2022), nov. 2022, doi: 10.15405/epsbs.2022.11.41.
- [51] H. Jing et Y. Fan, « Digital Transformation, Supply Chain Integration and Supply Chain Performance: Evidence From Chinese Manufacturing Listed Firms », *SAGE Open*, vol. 14, n° 3, p. 21582440241281616, juill. 2024, doi: 10.1177/21582440241281616.
- [52] M. Ai, Y. Yu, et Y. Bu, « The impact of digital transformation of resource-based enterprises on green innovation: Mechanism analysis based on TOE framework », *Innovation and Green Development*, vol. 4, n° 4, p. 100262, août 2025, doi: 10.1016/j.igd.2025.100262.
- [53] A. De Carolis, M. Macchi, E. Negri, et S. Terzi, « A Maturity Model for Assessing the Digital Readiness of Manufacturing Companies », in *Advances in Production Management Systems. The Path to Intelligent, Collaborative and Sustainable Manufacturing*, H. Lödding, R. Riedel, K.-D. Thoben, G. von Cieminski, et D. Kiritsis, Éd., Cham: Springer International Publishing, 2017, p. 13-20. doi: 10.1007/978-3-319-66923-6_2.
- [54] S. Mittal, M. A. Khan, D. Romero, et T. Wuest, « A critical review of smart manufacturing & Industry 4.0 maturity models: Implications for small and medium-sized enterprises (SMEs) », *Journal of Manufacturing Systems*, vol. 49, p. 194-214, oct. 2018, doi: 10.1016/j.jmsy.2018.10.005.
- [55] B. Bisht et al., « Industry 4.0 digital transformation: Shaping the future of food quality », *Food Control*, vol. 170, p. 111030, avr. 2025, doi: 10.1016/j.foodcont.2024.111030.
- [56] M. Mahin, N. Kadasah, A. Alsabban, et S. Albliwi, « Exploring the landscape of quality 4.0: a comprehensive review of its benefits, challenges, and critical success factors », *Production & Manufacturing Research*, vol. 12, n° 1, p. 2373739, déc. 2024, doi: 10.1080/21693277.2024.2373739.